

tion for various applications. Polaris integrates with Septentrio real-time kinematics (RTK) receivers, allowing for rapid attainment of centimetre-level accuracy. With expanded coverage in Western Europe, Polaris offers an extended geographical reach and enhanced capabilities. Point One's FusionEngine software complements the technology by seamlessly integrating cameras and additional sensors, ensuring the desired precision even in the absence of satellite signals. The employed GNSS receivers incorporate advanced interference mitigation (AIM+) technology, effectively suppressing a wide range of interference sources, including narrowband and wideband jammers. Polaris offers customisable performance and pricing options to meet the specific needs of individual applications, making it a versatile solution for industries such as industrial autonomy, precision agriculture, logistics, and autonomous vehicles.

Point One Navigation, <https://pointonenav.com>

Septentrio, <https://www.septentrio.com>

BLE based asset tracker

Onsemi has launched an end-to-end positioning system powered by its RSL15 microcontroller unit (MCU), offering accurate, cost-effective, and energy-efficient asset tracking solutions. The RSL15 MCU is the industry's most power-efficient Bluetooth 5.2 MCU, catering to industrial and medical sectors with connected smart devices. The MCU's integrated power management, versatile GPIO, clocking scheme, and array of peripherals provide design flexibility for high performance and energy-efficient applications. The system is suitable for diverse industrial automation applications, including asset tracking, smart retail, and IoT edge nodes. The MCU incorporates 80kb RAM and is availa-



ble with 284kb or 512kb flash options, enabling extended range, higher data transmissions, and precise localisation capabilities using AoA and AoD techniques. The MCU's smart sensing feature enhances power efficiency by allowing the Cortex M33 processor to enter deep sleep mode while monitoring sensor interfaces. Security is reinforced with Arm TrustZone and Arm CryptoCell-312 technology, ensuring authenticity, integrity, and confidentiality of code and data.

Onsemi

<https://www.onsemi.com>

Digital absolute pressure sensor

Infineon Technologies has launched two XENSIV barometric air pressure (BAP) sensors, the KP464 and KP466, catering to automotive applications.

The KP464 sensor is designed for



engine control management, optimising air-fuel mixture and improving fuel efficiency, leading to reduced power

consumption and lower emissions. It combines density measurement with manifold air pressure monitoring, resulting in lower power consumption and enhanced device productivity. In contrast, the KP466 sensor targets seat comfort functions, enabling pneumatic seat systems with features such as multi-contour seating, adjustable lumbar support, massage functions, and dynamic driving adjustments for added passenger comfort. Both are high-performance, high-precision digital absolute pressure sensors with SPI interfaces, integrating temperature sensors and diagnostic functions. They fulfill the requirements of the automotive industry and can be integrated in a daisy chain configuration with other Infineon system components, reducing connection pins and overall power consumption. The KP464 sensor enhances fuel efficiency and reduces

emissions, while the KP466 sensor enables innovative and customisable seat comfort features for a more pleasant driving experience.

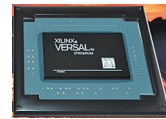
Infineon Technologies

<https://www.infineon.com>

Adaptive SoC for prototyping

AMD Versal Premium VP1902 adaptive system-on-chip (SoC) is claimed to be the world's largest adaptive SoC for emulation and prototyping.

With twice the capacity of previous-generation FPGAs, the VP1902 SoC aims to streamline the verification process



for complex semiconductor designs, enabling chipmakers to validate ASICs and SoC designs more efficiently and bring next-generation technologies to market faster. Boasting 18.5 million logic cells and twice the programmable logic density of its predecessor, the Virtex UltraScale+ VU19P FPGA, the VP1902 SoC offers industry leading capacity and connectivity. It also provides four times the aggregate I/O bandwidth, making it an ideal choice for emulating and prototyping AI, autonomous vehicles, Industry 5.0, and other emerging technologies. The VP1902 SoC offers great debugging capabilities, enabling faster iteration and bug fixing for chip designers.

AMD

<https://www.amd.com>

BOARDS & MODULES

Medical grade DC/DC converter

RECOM's REM60-W is a high-power, medical-grade DC/DC converter offering good performance, compact size,



and advanced protection features. The REM60-W is a cost-effective solution available in two input ranges: 9-36V DC and 18-75V DC. The module comes in a compact size measuring just